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Software Engineering for Internet of Things
An IoT Based Monitoring and Alert System for
Nuclear Power Plants

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1. Introduction

Nuclear energy is a reliable and low-carbon source of electricity that helps meet global energy demands. However, the safe operation of nuclear power plants is critical, as even minor malfunctions in systems such as the coolant circuit or pressure control can lead to serious consequences. This concern arises from potential risks such as reactor malfunctions, coolant system failures, and other safety-critical system failures. Continuous monitoring of critical parameters and providing timely alerts to plant operators are therefore essential to prevent accidents, maintain operational safety, and protect both the environment and the public.

In this project, we developed an Internet of Things (IoT)-based nuclear power plant monitoring system that continuously observes key reactor parameters. The system integrates data from multiple virtual sensors, applies data analysis techniques, and implements automated alert mechanisms.

The motivation behind this project is to explore how IoT technologies can be applied to nuclear power plant monitoring. By simulating realistic reactor parameters, monitoring critical system variables, and integrating automated alerting when abnormal, this project demonstrates a practical approach to enhancing operational safety, supporting decision-making and understanding the challenges of monitoring safety-critical systems.

2. Objectives

The primary objective of this project is:

1. Develop an IoT-based monitoring system capable of continuously observing critical reactor coolant system parameters, including pressure, coolant temperature, water level, reactor power, hydrogen concentration, and radiation levels in real time.
2. Enable reliable and efficient communication between IoT sensors and system components for continuous data transmission.
3. Detect abnormal variations in sensor values that may indicate unsafe operating conditions or potential accidents in the nuclear reactor.
4. Integrate an automated alerting mechanism to instantly notify plant operators when abnormal behavior is detected.
5. Store all sensor data in a time-series database to support both real-time monitoring and historical data analysis.
6. Provide a user-friendly, web-based dashboard that displays live sensor measurements and visualizes trends over time.

7. Demonstrate the application of IoT technologies in safety-critical environments, using a nuclear power plant monitoring scenario as an educational and experimental case study.

3. Functional Requirements

ID	Name	Description
FR-1	Data Generation	The system must be able to generate simulated data from virtual sensors representing various parameters of the reactor coolant system.
FR-2	Data Collection	The system should collect and store data from all virtual sensors deployed in the reactor coolant system in real-time.
FR-3	Accident Detection	The system should be able to analyze sensor data to detect potential loss-of-coolant accidents (LOCAs) in the reactor coolant system and trigger alerts.
FR-4	Monitoring	The system should be able to continuously monitor all sensor readings that provide information on the plant's operational status.
FR-5	Dashboard	The system must interactive dashboards that visualize both real-time and historical sensor data using graphs.
FR-6	Notification	The system should provide early notifications to operators when a potential accident or abnormal condition is detected on the plant.

Table 1: Functional Requirements

4. Non-Functional Requirements

ID	Name	Description
NFR-1	Scalability	The system should allow the addition of new virtual sensors without requiring changes to the simulator source code, supporting seamless expansion of the nuclear plant monitoring system.
NFR-2	Reliability	The system should operate continuously with minimal downtime.

NFR-3	Response	The system should provide alerts within a few seconds of detecting a potential loss-of-coolant accident or abnormal condition.
NFR-4	Usability	The system should provide a user interface with clear visualizations, enabling operators to easily monitor and navigate real-time sensor data.
NFR-5	Security	The system should support access control mechanisms to prevent unauthorized access and ensure data confidentiality.

Table 2: Functional Requirements

5. System Architecture

The architecture of our IoT-based nuclear power plant monitoring and accident detection system is composed of sensor simulation, a message broker, a gateway layer, a time-series database, visualization dashboards, and an alerting mechanism.

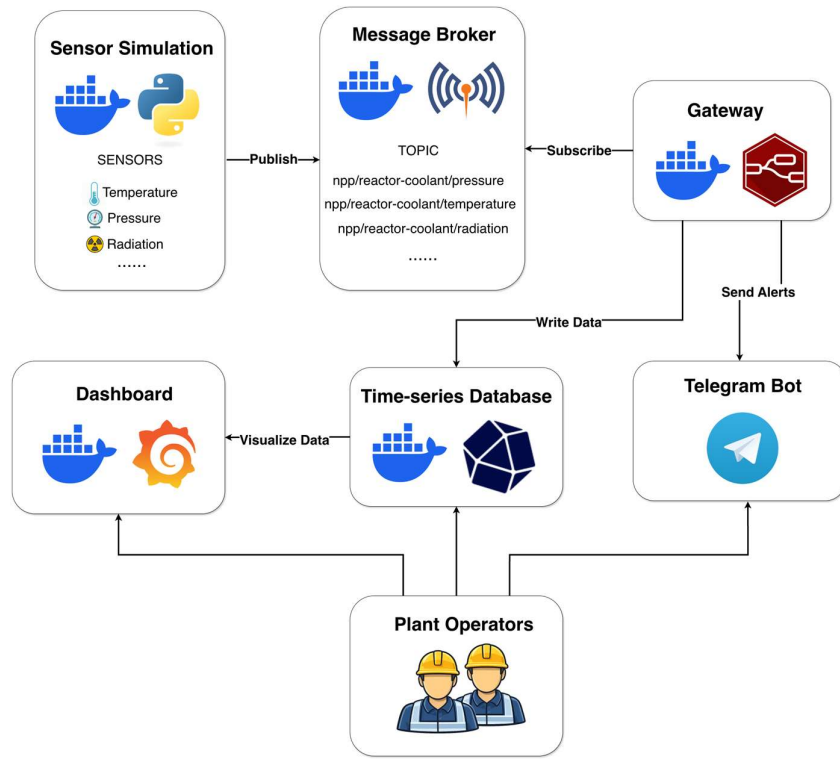


Figure 1. System architecture

The system simulates virtual sensors within the reactor coolant system, including temperature, pressure, radiation, water level, and other safety-critical variables, using Python. These virtual sensors generate realistic reactor coolant system values over time by leveraging the Nuclear Power Plant Accident Data (NPPAD) dataset.

The MQTT protocol is used as the communication protocol to publish simulated sensor data to a message broker. Each sensor publishes its data to a dedicated topic following the structure:

npp/reactor-coolant/<sensor-data>

The transmitted sensor data includes temperature, pressure, radiation levels, water level, reactor power, radiation dose, and hydrogen concentration.

A gateway layer is implemented using Node-RED, which subscribes to the relevant MQTT topics. This gateway acts as the central processing component responsible for:

- Analyzing incoming sensor data to identify abnormal operating conditions,
- Triggering alert notifications when necessary, and
- Storing all incoming measurements to a time-series database.

The alerting mechanism is implemented in Node-RED by evaluating sensor readings against predefined safety thresholds, such as elevated radiation levels or sudden pressure drops. When an abnormal condition is detected, the system sends an alert to the plant operator via Telegram, including information about the detected potential accident.

All sensor measurements are stored in an InfluxDB time-series database along with timestamps, enabling both real-time monitoring and historical analysis.

Finally, the stored data is retrieved from the time-series database and visualized using Grafana dashboards. These dashboards present real-time and historical sensor data through charts and graphs, and allow plant operators to observe live reactor conditions, identify trends over time, and maintain situational awareness of the reactor coolant system.

6. Dataset Used

Nuclear Power Plant Accident Data (NPPAD)

For our simulator data, we used the **Nuclear Power Plant Accident Data (NPPAD)** dataset, a publicly available time-series dataset that covers a wide range of operating conditions and accident scenarios for pressurized water reactor (PWR) nuclear power

plants. The dataset was generated using **PCTRAN**, a validated reactor transient and accident simulation software, and includes time-series records of multiple subsystem parameters under both normal and abnormal operations. NPPAD provides detailed measurements of reactor behavior during events such as loss-of-coolant accidents, steam line breaks, and other fault conditions, along with corresponding severity information and subsystem actions. This comprehensive dataset enables realistic simulation of reactor conditions and supports research and development of monitoring, diagnosis, and decision-making methods for safety-critical nuclear systems.

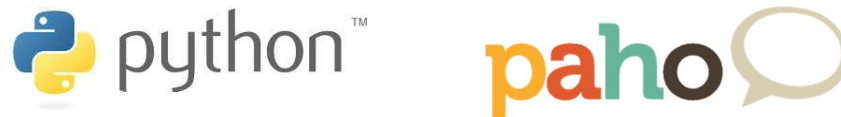
7. Technologies Used

The following technologies were used for the development of this project:

7.1. Python Paho MQTT

Python is used to simulate different virtual IOT sensors to generate synthetic data. The simulated sensors are temperature, pressure, radiation levels, water level, reactor power, radiation dose, and hydrogen concentration.

While paho-mqtt is used as an MQTT client to publish sensor measurements data to the MQTT broker. The data are published to the topics following the structure: npp/reactor-coolant/<sensor-data>



7.2. Mosquitto - MQTT Broker

Mosquitto is used as the MQTT broker for message exchange within the system. It enables lightweight, reliable, and asynchronous communication between the simulated sensors and the components. We used message broker to subscribe to the relevant MQTT topics to receive sensor data in real time.



7.3. Node-RED and Telegram

Node-RED is used for analyzing sensor data and storing in InfluxDB. This analysis involves evaluating key parameters against predefined thresholds to detect abnormal behaviour in reactor. When abnormal situations such as radiation increases or pressure drop are detected, Node-RED triggers notifications that are sent to the plant operators

via Telegram. Telegram bot is used to deliver alert messages containing information about the pressure, radiation, and detected accident.



Our Node-Red Flow is as follows:

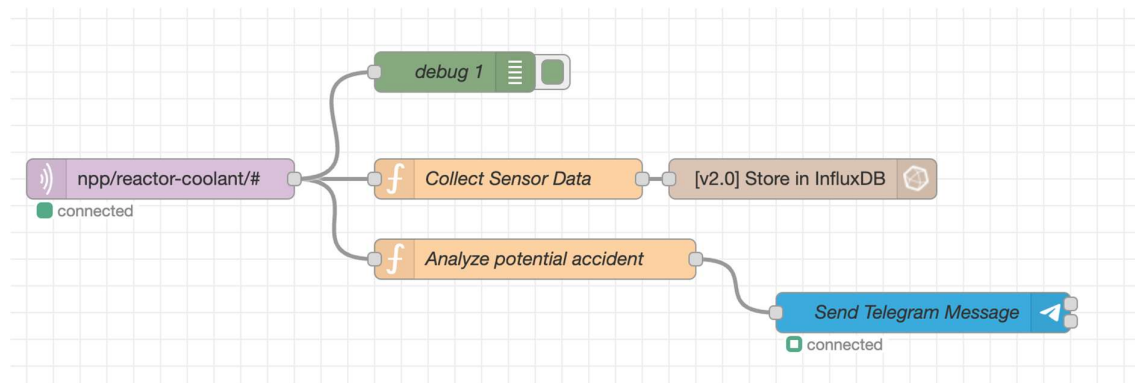


Figure 2. Node-Red flow that analyzes the information and write to InfluxDB and alert through the telegram bot

7.4. InfluxDB

InfluxDB is used as the time-series database for storing all sensor measurements. It is optimized for high write throughput and efficient querying of time-stamped data, making it ideal for IoT sensor data storage, and real-time analytics.



7.5. Grafana

Grafana is used to visualize the sensor data stored in InfluxDB. Interactive dashboards are created to display real-time and historical sensor values using charts and graphs.



A sample dashboard is shown below:

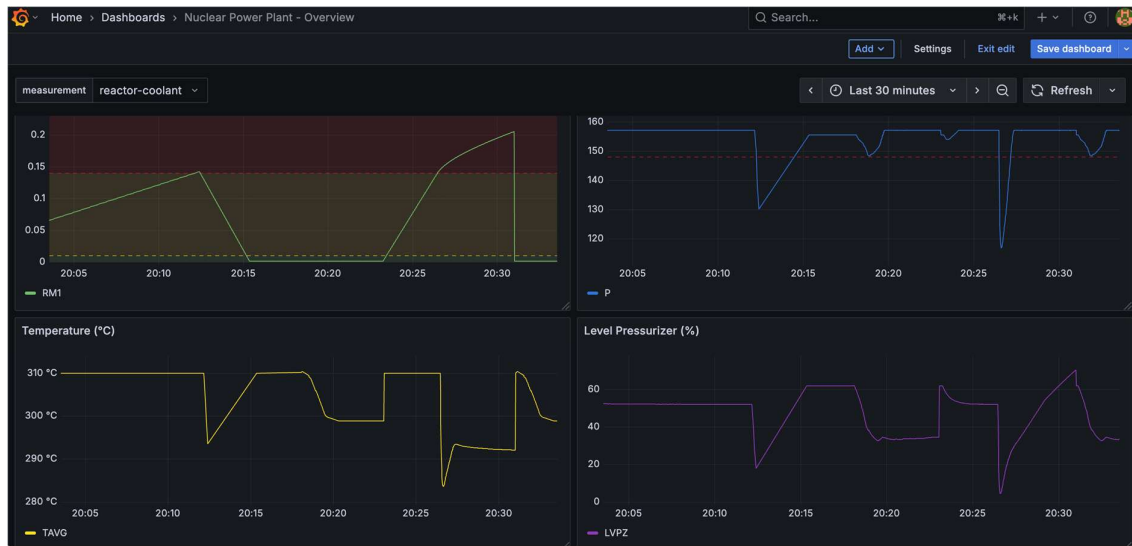


Figure 3. Data visualization in Grafana Dashboard

7.6. Docker Compose

Docker Compose is used for fully containerizing the system to ensure consistent deployment and execution across different environments and machines.

8. Conclusion

In this project, we designed and implemented a nuclear power plant IOT system that demonstrates how sensor data can be generated, transmitted, stored, visualized, and monitored in an integrated environment. The system shows how modular and containerized components can be combined to build a scalable and practical smart nuclear power plant solution with real-time monitoring and alerting capabilities.